

RIMINI, Italy

WMO: 161490

Lat: 44.0244N

Lon: 12.6128E

Elev: 43

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.4	28.7	19.1	14.4	36.4	22.7	17.1	35.2	23.2	41.2	21.1	41.2	4.2	200	0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	89.8	75.9	87.7	76.4	85.8	75.7	81.5	84.9	79.5	84.0	77.7	83.0	9.0	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.7	159.5	83.9	78.5	148.1	82.8	75.8	135.0	81.3	45.3	85.6	43.1	84.3	41.1	83.3	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.8	16.8	14.6		DB	21.4	95.4	4.9	3.4	17.9	97.9	15.0	99.8	12.3	101.7	8.8	104.2
				WB	20.8	84.3	4.9	2.3	17.3	85.9	14.5	87.3	11.7	88.6	8.2	90.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.8	40.6	42.2	48.4	54.7	63.2	71.2	75.6	75.3	67.6	60.0	50.9	42.2
	DBStd	13.52	5.18	5.43	5.68	4.78	4.94	5.13	4.39	4.32	5.04	5.17	6.44	5.76
	HDD50	944	296	222	96	13	0	0	0	0	2	64	251	
	HDD65	3650	757	637	515	309	95	9	0	0	29	168	423	708
	CDD50	3778	4	5	46	155	409	637	795	786	528	313	92	8
	CDD65	1009	0	0	0	1	39	196	330	321	107	14	1	0
	CDH74	9191	0	0	1	5	272	1721	3428	3097	630	36	1	0
	CDH80	2689	0	0	0	1	43	421	1148	982	93	1	0	0
(14) Wind	WSAvg	5.7	5.6	5.8	6.1	5.8	5.5	5.8	5.9	5.5	5.6	5.0	5.5	5.8
(15) Precipitation	PrecAvg	28.2	1.5	1.7	1.9	2.3	1.9	2.2	2.3	2.6	2.5	3.5	3.2	2.4
	PrecMax	45.0	4.1	4.9	8.5	6.5	7.0	6.2	9.1	8.9	8.2	12.5	6.9	4.5
	PrecMin	17.4	0.0	0.0	0.2	0.6	0.0	0.6	0.2	0.1	0.3	0.1	0.1	0.0
	PrecStd	6.0	1.2	1.2	1.6	1.5	1.6	1.3	2.0	2.2	1.7	3.1	1.9	1.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.8	61.2	68.9	73.2	84.3	89.9	93.6	93.5	86.2	77.9	70.2	62.9
		MCWB	53.7	53.0	57.1	60.8	69.8	74.6	73.7	74.4	73.3	68.5	62.6	56.5
	2%	DB	55.7	57.5	64.7	69.6	80.4	86.4	90.0	89.6	82.5	73.8	66.4	57.5
		MCWB	49.8	50.4	54.7	60.6	68.6	74.6	75.5	77.1	74.4	67.5	61.0	52.2
	5%	DB	51.9	54.6	61.1	66.9	76.9	84.4	87.9	87.5	80.3	71.6	63.2	53.7
		MCWB	47.2	48.9	53.6	59.0	67.4	73.8	76.1	77.3	72.7	66.1	59.5	50.7
	10%	DB	48.5	51.8	58.8	64.6	73.8	82.3	85.9	85.4	77.3	69.4	60.8	51.1
		MCWB	45.8	47.4	52.6	58.0	65.9	73.0	75.9	76.5	70.0	64.8	57.6	48.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.5	54.8	59.1	64.8	74.7	81.0	83.9	85.4	80.7	71.5	64.9	57.6
		MCDB	59.4	59.4	65.5	68.5	78.3	84.6	86.1	87.0	82.6	75.0	67.2	61.0
	2%	WB	51.0	51.9	56.6	62.8	71.5	78.7	81.3	81.8	76.7	69.1	62.2	53.8
		MCDB	54.3	55.4	61.7	66.7	76.9	82.9	85.1	85.1	80.3	72.3	65.0	56.4
	5%	WB	48.4	50.0	55.1	60.9	69.0	76.5	79.4	79.6	73.6	67.2	60.2	51.3
		MCDB	50.8	53.1	59.3	65.0	74.4	81.7	84.3	84.1	77.8	70.3	62.8	53.4
	10%	WB	46.4	48.2	53.4	59.2	66.9	74.5	77.6	77.6	71.4	65.3	58.2	49.0
		MCDB	48.4	51.2	57.8	63.3	72.5	80.3	83.2	83.2	76.2	68.3	60.6	50.6
(35) Mean Daily Temperature Range	5% DB	MDBR	12.4	14.4	15.6	15.6	16.7	16.6	16.7	16.5	15.9	13.9	12.2	12.3
		MCDBR	17.8	18.7	19.9	19.3	20.8	19.0	19.4	18.7	17.7	16.6	14.8	16.1
		MCWBR	12.9	12.8	12.4	12.5	12.8	12.1	12.9	13.3	11.6	11.2	10.6	12.0
	5% WB	MCDBR	15.6	15.7	17.6	16.9	19.2	17.1	16.8	17.1	16.0	15.3	12.9	14.1
		MCWBR	12.2	11.7	12.4	12.4	13.5	12.6	13.2	13.9	12.4	11.5	10.3	11.4
(40) Clear-Sky Solar Irradiance	taub		0.357	0.377	0.411	0.449	0.447	0.452	0.443	0.438	0.425	0.411	0.380	0.352
	taud		2.393	2.311	2.231	2.146	2.204	2.239	2.267	2.298	2.308	2.324	2.338	2.421
	Ebn at Noon		239	253	259	259	263	262	262	259	252	238	225	228
	Edh at Noon		25	32	39	46	45	43	42	39	36	31	26	23
(44) All-Sky Solar Radiation	RadAvg		431	699	1104	1476	1838	2096	2147	1841	1331	844	489	385
	RadStd		64	102	142	135	151	107	97	120	91	87	55	51

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.85	N/A	N/A	N/A	N/A	N/A	-140	+148	N/A
(47) Regional (17 neighbors)		+0.72	N/A	N/A	N/A	N/A	N/A	-155	+139	N/A

Nomenclature: See separate page